Modülasyon

$BG\{m(t)\} = B$ Hz $M(f) = 0,\ |f| > B$

$c(t) = A_c \cos(2\pi f_c t + \theta_c)$

* Genlik modülasyonu (Amplitude modulation - Am)
* Frekans " (Frequency modulation - Fm) } Lineer olmayan (nonlineer) üstel (exponential) mod
* Faz " (Phase " - Pm) }

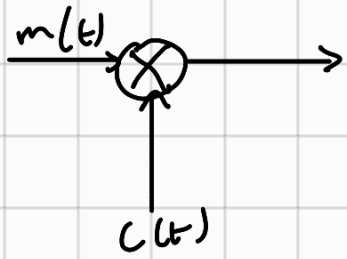
Double Sideband Suppressed Carrier AM (DSB-SC-AM)

Taşıyıcının Bastırılmış (TBB) çift Yan Bandlı (ÇYB)
Genlik modülasyonu (Gm)

$m(t)$: Mesaj haber işareti / modüle eden işaret (modulating signal)

$c(t) = A_c \cos(2\pi f_c t)$ Taşıyıcı işaret (carrier signal)

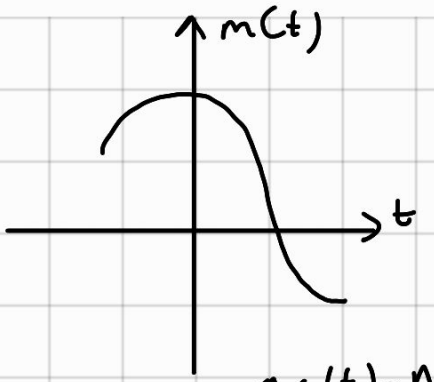
$y(t) = m(t)c(t) = A_c m(t) \cos(2\pi f_c t)$
↳ modüle edilmiş işaret (modüle edilmiş işaret) modulate signal

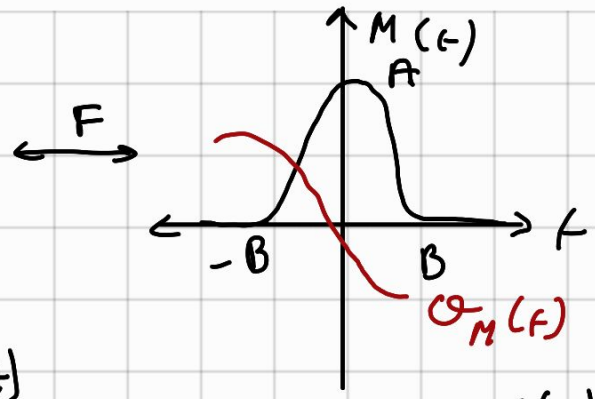


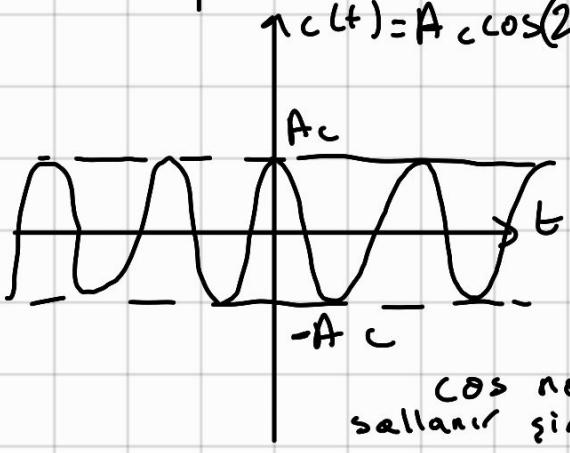
$g(t) = A_c m(t) \cos(2\pi f_c t)$

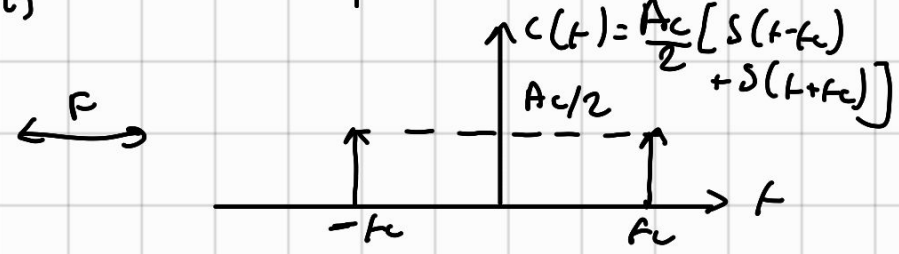
$\Updownarrow$

$Y(f) = \frac{A_c}{2}\left[M(f-f_c) + M(f+f_c)\right]$

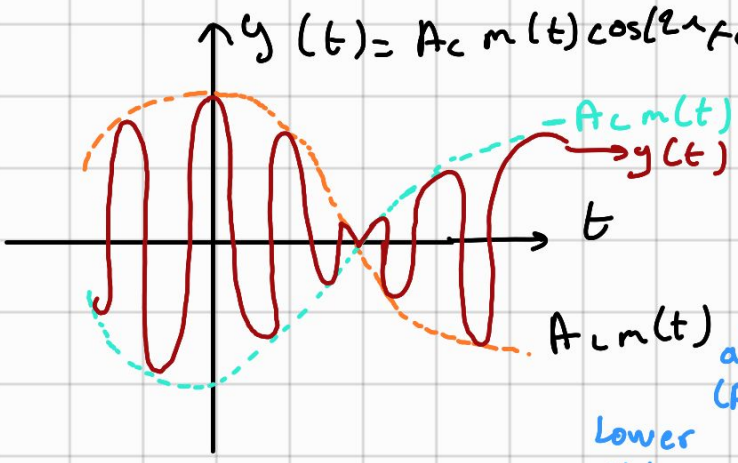


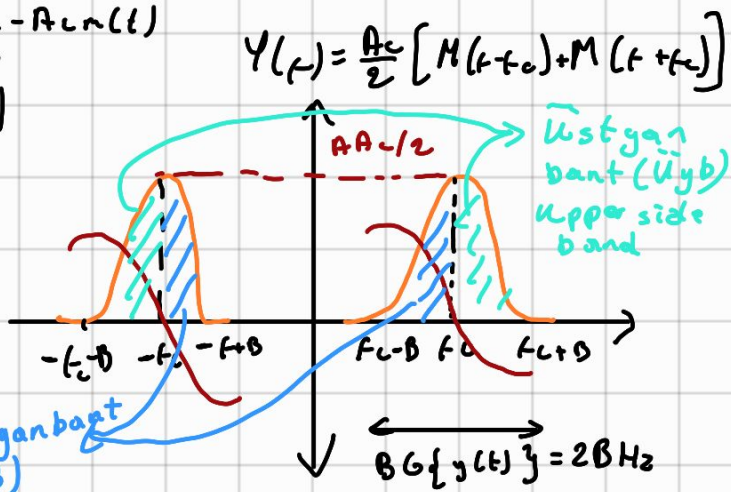






cos normalde -1 ile 1 arasında salınır şimdi Acm(t) ile -Acm(t) arasında salınır.





alt yanbant (AYB)

Lower side band ♡

Modülasyon işlemi sonucunda bant genişliği ikiye katlanıyor. (işaret bant geçiren dönüşür)

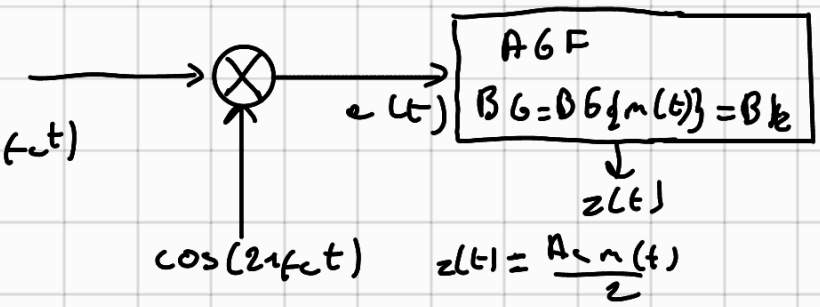
Mesaj işaretleri temel banttadır hep zaman

Taşıyıcısı bastırılmış

Çünkü frekansa bakınca taşıyıcının frekans bileşenleri gözükmüyor

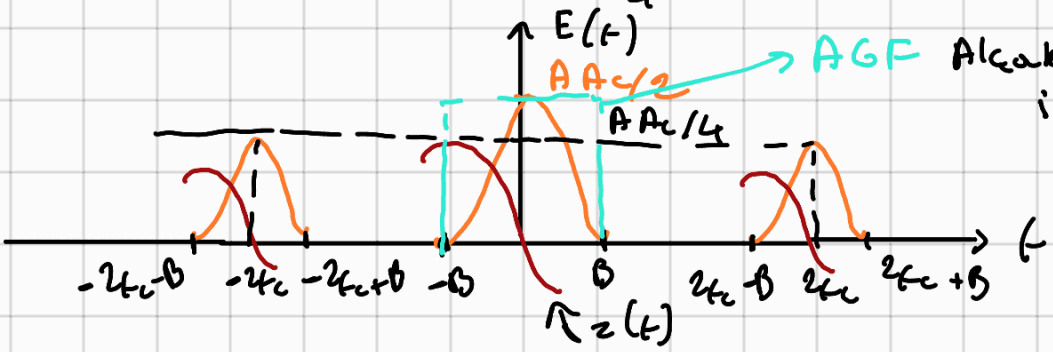
Demodülasyon

$$y(t) = m(t)\,c(t)$$
$$= A_c\, m(t) \cos(2\pi f_c t)$$



$$e(t) = y(t)\cos(2\pi f_c t) \rightarrow E(f) = \frac{1}{2}\left[Y(f-f_c) + Y(f+f_c)\right]$$
$$= A_c\, m(t)\cos^2(2\pi f_c t)$$
$$\downarrow \frac{1}{2}\left[1+\cos(4\pi f_c t)\right]$$
$$e(t) = \frac{A_c}{2} m(t) + \frac{A_c}{2} m(t)\cos(2\pi \cdot 2f_c \cdot t)$$

$$E(f) = \frac{A_c}{2} M(f) + \frac{A_c}{4}\left[M(f-2f_c) + M(f+2f_c)\right]$$



Alçak geçiren ile çarparsak işareti elde edebiliriz

$$Z(f) = \frac{A_c}{2} M(f) \rightarrow z(t) = \frac{A_c}{2} m(t)$$

fc yi küçük seçersek frekansta bileşenler birbirinin üzerine biner ve sinyali tekrar ayıramayız

$$2f_c - B \geq B$$
$$2f_c \geq 2B$$
$$f_c \geq B$$

Bir de $f_c - B \geq 0 \Rightarrow f_c \geq B$ Teorik

* Pratik olarak bir anten bir dar bandı işareti ($f_c >> B$) bozulmasız yayınlayabilir

$\frac{f_c}{B} >> 1$

Ticari genlik modülasyonunda (Radyo yayınları): 550 kHz - 1600 kHz

$$100 < \frac{f_c}{B} < 300$$

TB - ÇYB - GM'nin gücü

$$y(t) = m(t)\,c(t) = A_c m(t)\cos(2\pi f_c t)$$

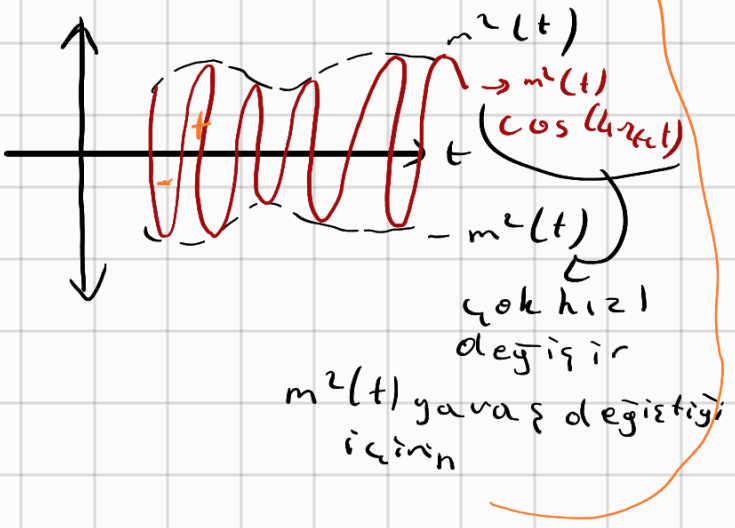
$$P_y = \lim_{T\to\infty} \frac{1}{T}\int_{-T/2}^{+T/2} y^2(t)\,dt = \lim_{T\to\infty}\frac{1}{T}\int_{-T/2}^{T/2} A_c^2 m^2(t)\cos^2(2\pi f_c t)\,dt$$

$\cos^2(2\pi f_c t) = \frac{1}{2}[1+\cos(4\pi f_c t)]$

$$= \frac{A_c^2}{2}\left[\lim_{T\to\infty}\frac{1}{T}\int_{-T/2}^{T/2} m^2(t)\,dt + \lim_{T\to\infty}\frac{1}{T}\int_{-T/2}^{+T/2} m^2(t)\cos(4\pi f_c t)\,dt\right]$$

$$= \frac{A_c^2}{2}$$

$$\boxed{P_y = \frac{A_c^2}{2} P_m}$$



Synchronus / Phase coherent demod

Senkron / Faz tahminli demodülasyon

1- Phase Locked Loop (PLL) devresi
(Faz kilitlemeli Döngü)

2- Pilot taşıyıcı kullanmak $y(t)$ ye $A_p \cos(2\pi f_c t)$ ($A_p \ll A_c$) ve bununla sinyal çarpılıp demode edilir → sayfa 133 Şekil 3.9 / 2.baskı 126

Örnek $m(t) = \text{sinc}(10^4 t)$

Find DSB-SC-AM signal and its BW if the carrier is a sinusoid with a frequency of 1MHz

$$y(t) = m(t)c(t)$$

$$c(t) = 1\cos(2\pi \cdot 10^6 t)$$

$$y(t) = \text{sinc}(10^4 t)\cos(2\pi \cdot 10^6 t)$$

$$A\,2\,\text{sinc}(2t) \overset{F}{\leftrightarrow} A\,\Pi(f/2)$$

$$\text{sinc}(10^4 t) \leftrightarrow M(f) = \frac{1}{10^4}\,\Pi\left(\frac{f}{10^4}\right)$$

$$B = \frac{10^4}{2} = 5\text{kHz}$$

$$BW\{y(t)\} = 2B = 10\text{kHz}$$

saf cosinüslü

Örnek tone modulation : $m(t) = A\cos(2\pi f_m t)$, $f_m < f_c$
çünkü tek bir frekansımız

$$M(f) = \frac{A}{2}\left[\delta(f-f_m) + \delta(f+f_m)\right]$$

$$y(t) = m(t)\,c(t) = m(t)\,A_c\cos(2\pi f_c t)$$
$$= A A_c \cos(2\pi f_m t)\cos(2\pi f_c t)$$
$$= \frac{A A_c}{2}\left[\cos(2\pi(\underbrace{f_c - f_m}_{=f_1})t) + \cos(2\pi(\underbrace{f_c+f_m}_{f_2})t)\right]$$

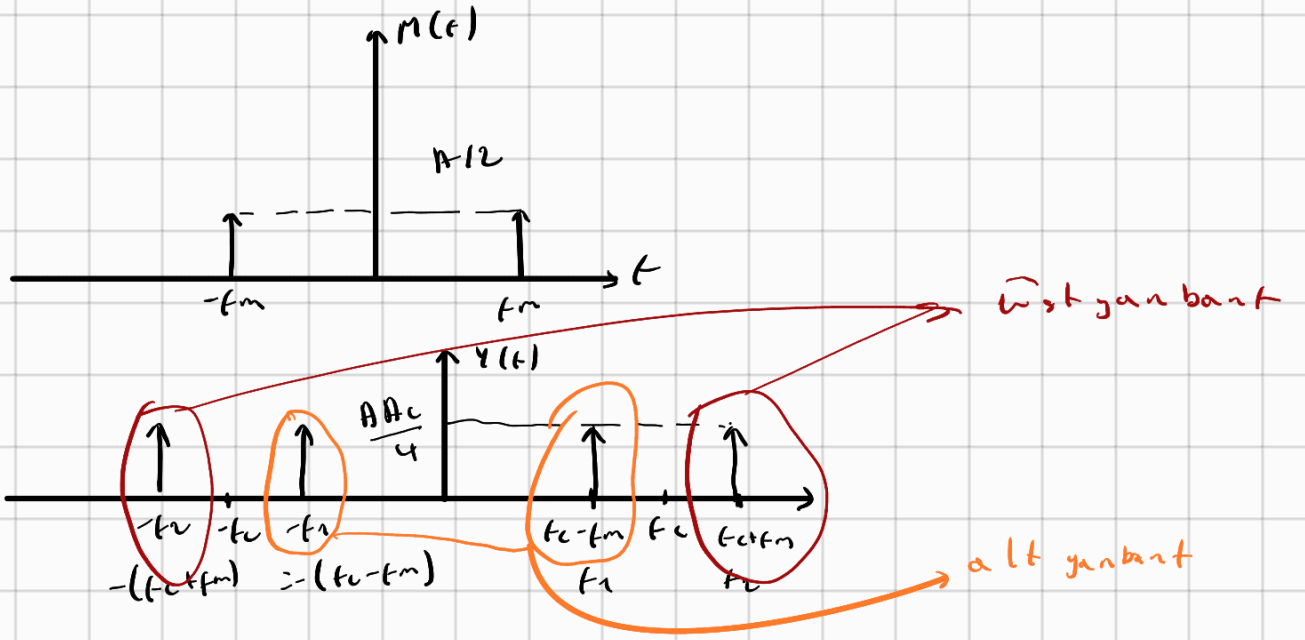
$$Y(f) = \frac{A A_c}{4}\left[\delta(f-f_1) + \delta(f+f_1) + \delta(f-f_2) + \delta(f+f_2)\right]$$

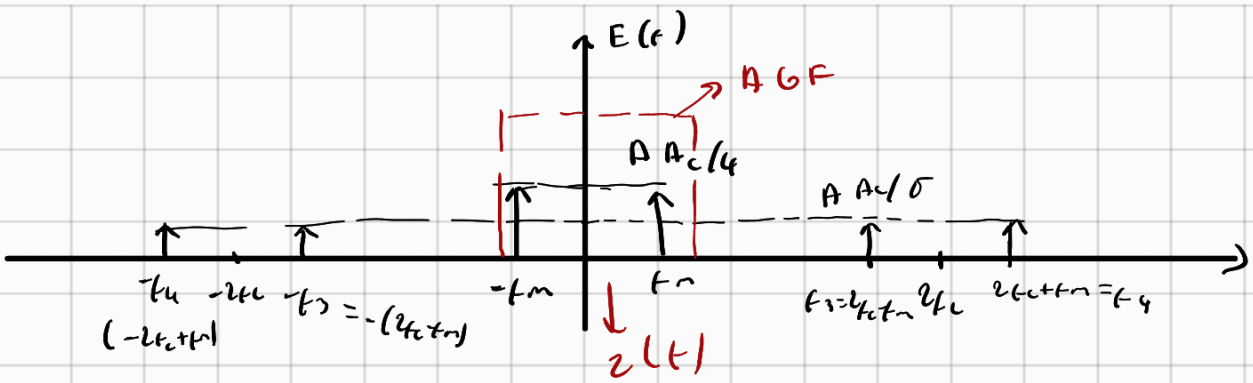
$$e(t) = y(t)\cos(2\pi f_c t) = A_c m(t) \underbrace{\cos^2(2\pi f_c t)}_{\frac{1}{2}\left[1+\cos(2\pi 2f_c t)\right]}$$
$$= \frac{A_c}{2} m(t) + \frac{A_c}{2} m(t)\cos(2\pi 2f_c t)$$
($A\cos(2\pi f_m t)$ ile $m(t)$ çarpımı)

$$e(t) = \frac{A A_c}{2}\cos(2\pi f_m t) + \frac{A A_c}{2}\underbrace{\cos(2\pi f_m t)\cos(2\pi 2f_c t)}$$

$$\frac{1}{2}\left[\cos(2\pi(\underbrace{2f_c - f_m}_{=f_3})t) + \cos(2\pi(\underbrace{2f_c+f_m}_{=f_4})t)\right]$$

$$E(f) = \frac{A A_c}{4}\left[\delta(f-f_m) + \delta(f+f_m)\right]$$
$$+ \frac{A A_c}{8}\left[\delta(f-f_3) + \delta(f+f_3) + \delta(f-f_4) + \delta(f+f_4)\right]$$





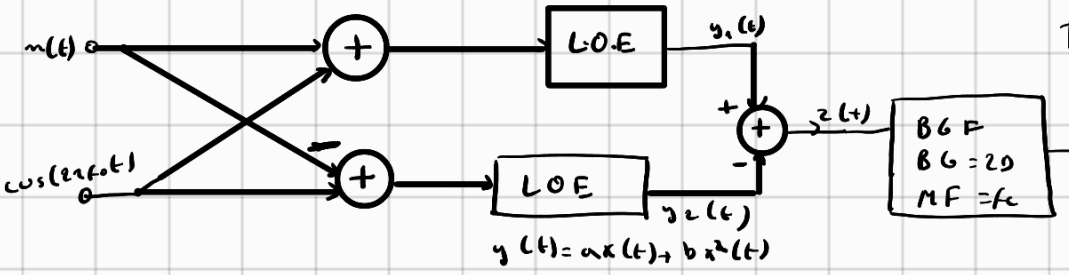
$$P_y = \frac{A_c^2}{2} P_n = \frac{A_c^2}{2} \cdot \frac{A^2}{2} = \frac{A^2 \cdot A_c^2}{4}$$

$$y_l(t) = \frac{A A_c}{2} \cos(2\pi f_1 t) \rightarrow P_{yl} = \frac{A^2 A_c^2}{8}$$

$$y_u(t) = \frac{A A_c}{2} \cos(2\pi f_2 t) \rightarrow P_{yu} = \frac{A^2 A_c^2}{8}$$

Modülatörler

- Multiplier modulator (Çarpıcı modülatör)
- Nonlinear modulator (Lineer olmayan modülatör)



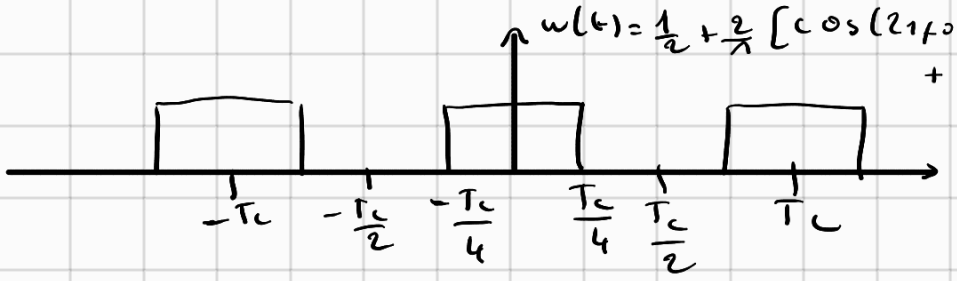
Single balanced modulator
Tek dengeli "

$4 b m(t) \cos(2\pi f_c t)$

girişinde taşıyıcı işareti olmadığı için

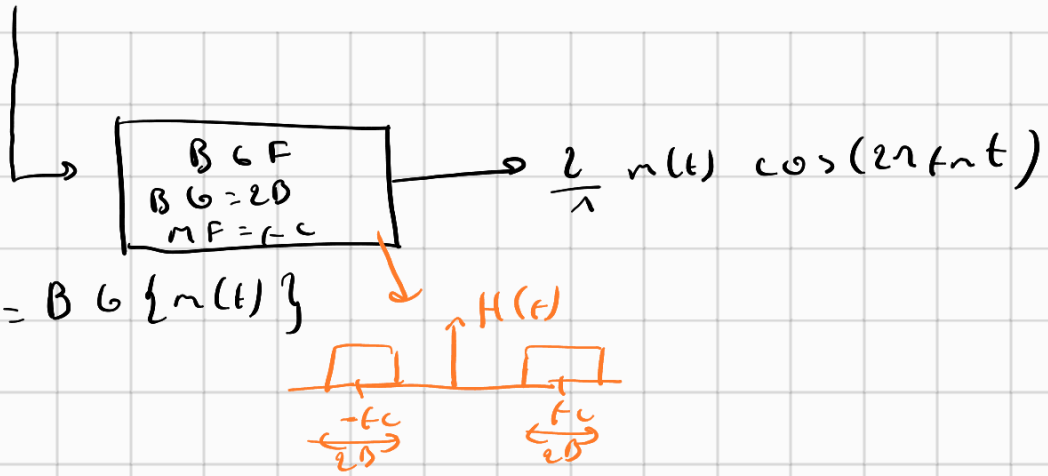
$$x_1(t) = m(t) + \cos(2\pi f_c t)$$
$$x_2(t) = -m(t) + \cos(2\pi f_c t)$$
$$z(t) = y_1(t) - y_2(t) = 2a m(t) + 4b m(t) \cos(2\pi f_c t)$$

-Anahtarlama modülatörü (Switching modulator)



$$w(t) = \frac{1}{2} + \frac{2}{\pi}\left[\cos(2\pi f_0 t) - \frac{1}{3}\cos(2\pi 3 f_c t) + \frac{1}{5}\cos(2\pi 5 f_c t) + \cdots\right]$$

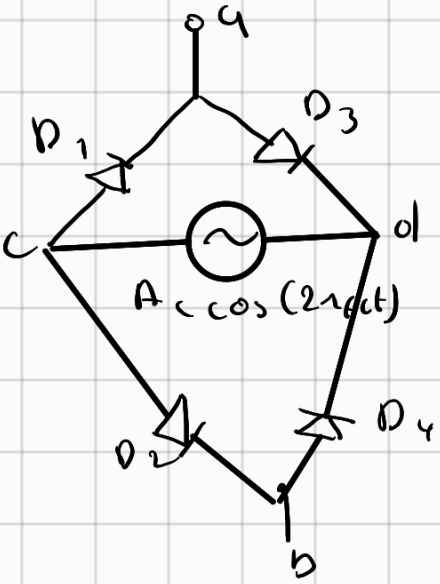
$$m(t) \cdot w(t) = \frac{1}{2} m(t) + \frac{2}{\pi}\left[m(t)\cos(2\pi f_c t) - \frac{1}{3} m(t)\cos(2\pi 3 f_c t) + \frac{1}{5} m(t)\cos(2\pi 5 f_c t) + \cdots\right]$$

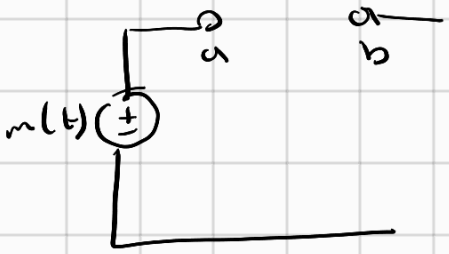


Nasıl çarpım yaparız



Shunt bridge modulator
(paralel köprülü modülatör)





Seri köprülü modülatör

$D_1 \equiv D_2$, $D_3 \equiv D_4$

$v_c > v_d \Rightarrow$ Diyotlar iletir

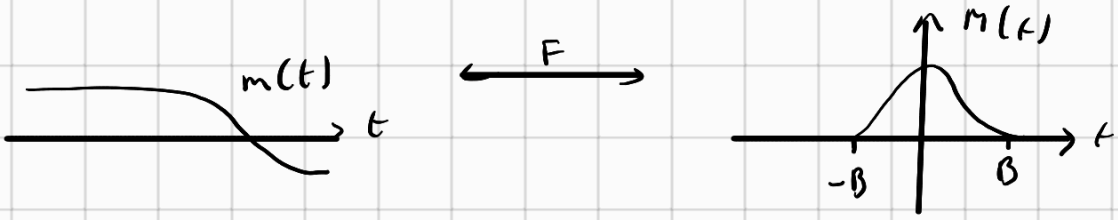
$\Rightarrow a = b$

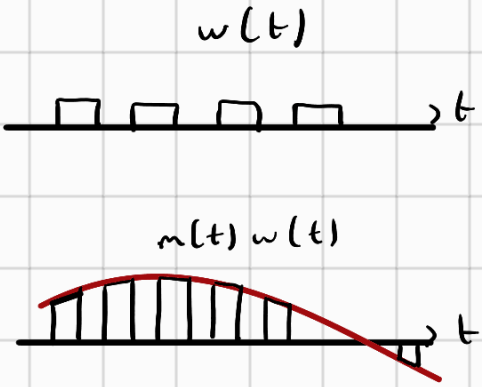
$V_a = V_b$ kısa devre

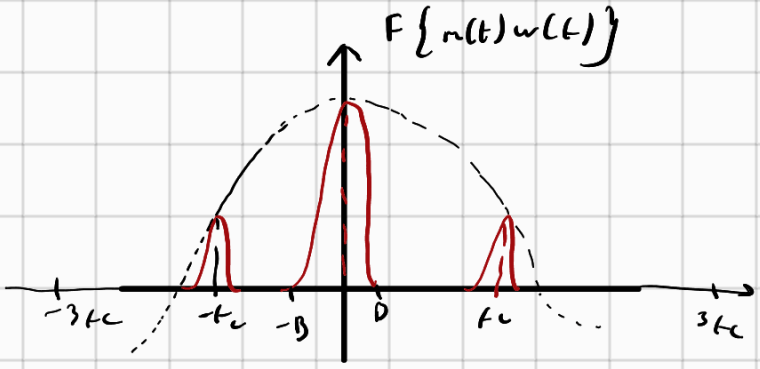
$v_c < v_d \Rightarrow$ Diyotlar kesimde

$\Rightarrow$ a ve b arası açık devre

Bu olay f_c ile periyodik olarak gerçekleşir

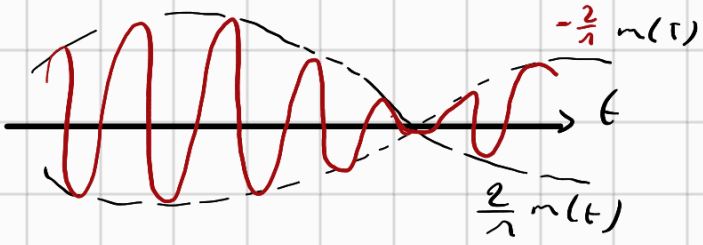






BG F
BG = 2B
MF = fc

$\rightarrow \frac{2}{\pi} m(t) \cos(2\pi f_c t)$



Ring modulator (Double balanced mod) okuyunuz *

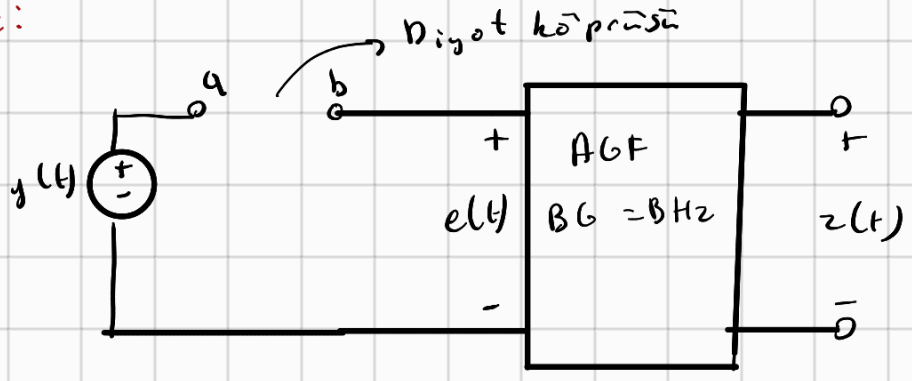
Demodülasyon olayını nasıl yapılır??

Demodülatör

Senkron demodülasyon yapılmalı

Çarpıcı + AGF

örnek:



$y(t) = A_c m(t) \cos(2\pi f_c t)$

$$e(t) = y(t)w(t)$$

$$= A_c m(t) \cos(2\pi f_c t) \cdot \underbrace{\left\{ \frac{1}{2} + \frac{2}{\pi}\left[\cos(2\pi f_c t) - \frac{1}{3}\cos(2\pi 3 f_c t) + \frac{1}{5}\cos(2\pi 5 f_c t) + \cdots \right] \right\}}_{w(t)}$$

$$= \underbrace{\frac{2A_c}{\pi} m(t)\cos^2(2\pi f_c t)}_{\frac{2A_c}{\pi} m(t)\frac{1}{2}\left[1+\cos(2\pi 2 f_c t)\right]} + \underbrace{\left(k m(t)\cos(2\pi f_c t)\right.}_{\text{genellştir}} \left. k m(t)\cos(2\pi n f_c t)\ n \neq 1\right)$$

$$= \frac{A_c}{\pi} m(t) + \frac{A_c}{\pi} m(t)\cos(2\pi 2 f_c t)$$

AG $\rightarrow$ $\frac{A_c}{\pi} m(t) = z(t)$